PIXELBIT

Pocket arcade: quick start

15 games + a video player + chat
No router and no computer needed to play.
Full guide: the README at github.com/Simplyph0/PixelBit

1. THE FOUR BUTTONS

The console has four buttons. These are all you need:

	UP	
LEFT	DOWN	RIGHT



The menu: LEFT / RIGHT scroll

- **Menu:** LEFT and RIGHT scroll through the games (hold to keep going). **UP** starts one.
- **DOWN** (tap) opens that game's settings. Games that have some show vSET. Hold DOWN for a second to replay the welcome tour.
- **In a game:** hold **UP + DOWN** together for a second to leave. On the result screen UP goes to the menu, RIGHT plays again. In **Chess**, LEFT / RIGHT pick a piece, UP selects it, LEFT / RIGHT pick the square, UP moves.
- **In a list** (SETTINGS, VIDEOS, CHAT): UP and DOWN scroll, **RIGHT** picks, **LEFT** goes back.

2. PLAY A FRIEND (TWO CONSOLES)

Pong, Tic-Tac-Toe, Rock Paper Scissors, Connect 4 and Chess. No router and no Wi-Fi: the two consoles talk to each other directly by radio. Keep them within a few metres, both on the same firmware version.

1. On **both** consoles pick the game, tap **DOWN** on the menu and choose **2 PLAYER**.
2. One console picks **HOST GAME**, the other picks **JOIN GAME**.
3. The joiner chooses the host's 5-letter ID with LEFT / RIGHT, then **UP**.
4. The host presses **UP** to accept (DOWN says no). Play!



Host or join



Joiner picks a host

3. CHAT

The **CHAT** tile (after VIDEOS) sends ready-made messages to other consoles over the same radio: "LET'S PLAY...", "REMATCH?", "GOOD GAME!" and more.

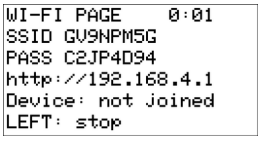


The CHAT tile

- **NEW MESSAGE:** pick who (everyone, a saved contact or a console nearby), then the message. **DELIVERED** = it arrived, **NO REPLY** = nobody answered.
- **INBOX** keeps the last 8. RIGHT answers one.
- A **round dot flashing in the top right corner** means a message is waiting, even in a game or a video.
- Your 5-letter ID is on the CHAT screen. Messages are not secret. Turn listening off: SETTINGS > CHAT > OFF.

4. WI-FI PAGE (BACKUP AND CONTACTS)

The console can make its **own private Wi-Fi network**, so a phone or computer can talk to it. It is for backing up scores and saving chat contacts.



What the console shows

1. **SETTINGS** (the last tile) > **WI-FI PAGE**. The screen shows a network name (SSID) and a random password: new every time.
2. Join that network with your phone or computer, then open `http://192.168.4.1`.
3. On the page: download or restore your scores, add chat contacts (a 5-letter ID and a name), or erase all scores.
4. Press **LEFT** on the console to switch it off. One device at a time.

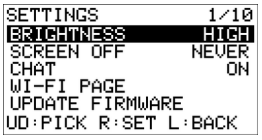
5. VIDEOS (NO SOUND)

The **VIDEOS** tile plays black and white clips. In the list RIGHT plays and LEFT leaves. While playing: **UP** pauses, LEFT / RIGHT jump 5 seconds, **DOWN** goes back to the list.

Add your own: pick **+ UPLOAD VIDEOS** and press RIGHT. It makes a Wi-Fi network just like in 4: join it, open `192.168.4.1`, upload a `.pbv` file, then press LEFT on the console. Make a `.pbv` on a computer with `tools/video/pbv.py` (for example `python3 pbv.py movie.mp4 --pixel 2`). A bigger `--pixel` is easier to see and makes a smaller file.

6. SETTINGS (THE LAST TILE)

BRIGHTNESS, **SCREEN OFF** (after 1, 2 or 5 minutes; any button wakes it), **CHAT** on/off, **WI-FI PAGE**, **UPDATE FIRMWARE**, resets, **SHOW INTRO** and **ABOUT** (version and your ID).



SETTINGS

Resets ask you to **hold RIGHT** (2 seconds, 3 for a factory reset), so a long press on the list cannot erase anything.

7. UPDATE ONE CONSOLE FROM ANOTHER

On both: **SETTINGS > UPDATE FIRMWARE**. One picks **RECEIVE**, the other **SEND** and finds it. The receiver sees what is coming: **UP** installs, **DOWN** says no. It takes about 10 seconds: do not switch off. If anything goes wrong the old firmware stays. Use two consoles wired the same way round (the button layout is copied too).

8. IF SOMETHING IS WRONG

- **Buttons feel mirrored:** you need the `-flipped` firmware file.
- **Blank screen:** check the display's two data wires (SDA and SCL).
- **Two consoles do not see each other:** get closer, host picks HOST GAME first, same firmware version.
- **An update sticks at 0%:** the receiving console has firmware older than v4.4: flash it once with a cable.

Stuck anywhere? Hold UP + DOWN for a second to get out of a game, back to the menu.